

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 678 Distributed Systems and Cloud Computing — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
MIT 6.5840 — Distributed Systems (free lectures and labs)	The primary source. Complete video lectures, the paper list, and the Raft labs. The best free distributed systems course that exists, and the labs are the real value.
Martin Kleppmann — Designing Data-Intensive Applications	The reference for Modules 03 through 08. Unusually careful about what the guarantees actually are. Library copy; the author's lecture series is free on YouTube.
van Steen & Tanenbaum — Distributed Systems (free book)	Free in full. Broader and more systematic than Kleppmann, stronger on naming, coordination, and the formal material.
Ongaro & Ousterhout — In Search of an Understandable Consensus Algorithm (free)	The Raft paper, and the Module 06 project. Deliberately written to be implementable, and the visualisation on the same site is excellent.
Gaffer On Games — networked physics and game networking (free)	The reference for Modules 09 through 11. Glenn Fiedler's series on prediction, reconciliation, and snapshot interpolation is the clearest treatment of netcode available.
Google — Site Reliability Engineering and the SRE Workbook (free)	The Module 13 material: SLOs, error budgets, tail latency, and what operating a distributed system actually involves.

Module-by-module map

Module 01 — Partial Failure

Resource	What it covers for this module
MIT 6.5840 — lecture 1, introduction (free)	The §1 framing and the course's overall shape. Watch it this week.
Deutsch & Gosling — The Eight Fallacies of Distributed Computing (free)	The §2 list, from the people who wrote it down. One page, and still entirely current.
Peter Bailis & Kyle Kingsbury — The Network Is Reliable (free)	Evidence for the first fallacy , collected from real production incidents. Read it before deciding your network is fine.
Kleppmann — Designing Data-Intensive Applications, chapter 8 ('The Trouble with Distributed Systems')	The best single treatment of §1 and §3 , including unreliable clocks and the process-pause problem.

Module 02 — The Network Underneath

Resource	What it covers for this module
Glenn Fiedler — UDP vs TCP, and Reliability over UDP (free)	The §1 argument , made concretely and with code. The whole series is worth reading.
Ilya Grigorik — High Performance Browser Networking (free book)	Free in full. The best treatment of §2 — BDP, congestion control, and why latency dominates — and strong on WebRTC for §3.
Gettys & Nichols — Bufferbloat: Dark Buffers in the Internet (free)	The §2 callout, from the people who identified and named the problem.
RFC 8445 (ICE), and Valve's GameNetworkingSockets (free)	The §3 traversal machinery, and a production implementation of the §4 recommendation with readable source.

Module 03 — Time, Clocks, and Ordering

Resource	What it covers for this module
Lamport — Time, Clocks, and the Ordering of Events in a Distributed System (free)	The paper that created §2. Short, readable, and one of the most cited papers in computing for good reason.
Kleppmann — Designing Data-Intensive Applications, chapter 8 ('Unreliable Clocks')	The §1 catalogue and the §4 warning , with real incident examples.
Kulkarni et al. — Logical Physical Clocks (HLC) (free)	The §3 default, from the paper that introduced it.
Corbett et al. — Spanner: Google's Globally-Distributed Database (free)	TrueTime and commit wait — the §1 callout, as shipped.

Module 04 — Replication and Consistency

Resource	What it covers for this module
Kleppmann — Designing Data-Intensive Applications, chapters 5, 6 and 9	The best treatment of this entire module , particularly the linearizability-versus-serialisability distinction of §2.
Herlihy & Wing — Linearizability: A Correctness Condition for Concurrent Objects (free)	The original definition. Worth reading for how carefully it is stated.
Terry et al. — Session Guarantees for Weakly Consistent Replicated Data (free)	The §3 result , and the paper that named read-your-writes and monotonic reads.
DeCandia et al. — Dynamo: Amazon's Highly Available Key-value Store (free)	The quorum system of §1, with the sibling-value conflict handling that LWW replaced in many later systems.

Module 05 — The Impossibility Results

Resource	What it covers for this module
Gilbert & Lynch — Brewer's Conjecture and the Feasibility of Consistent, Available, Partition-Tolerant Web Services (free)	The actual proof , with the actual definitions. Read the definitions section at minimum.
Brewer — CAP Twelve Years Later: How the Rules Have Changed (free)	The author correcting the folklore , which is the most persuasive item in §1.
Fischer, Lynch & Paterson — Impossibility of Distributed Consensus with One Faulty Process (free)	FLP itself. Short, and the proof idea is followable.
Abadi — Consistency Tradeoffs in Modern Distributed Database System Design (PACELC) (free)	The §3 framing , with real systems classified.

Module 06 — Consensus

Resource	What it covers for this module
Ongaro & Ousterhout — In Search of an Understandable Consensus Algorithm (free)	The Raft paper . Deliberately written to be implementable, and the interactive visualisation on the same site is excellent.
MIT 6.5840 — the Raft labs (free)	The best way to learn this is to implement it , and these labs have a test suite that finds the bugs you will have.
Lamport — Paxos Made Simple (free)	The other algorithm, from its author. Worth reading after Raft for the contrast in exposition.
Kleppmann — How to do distributed locking (free)	The fencing token argument of §3 , with the process-pause scenario worked through.

Module 07 — Failure Detection and Membership

Resource	What it covers for this module
Das, Gupta & Motivala — SWIM: Scalable Weakly-consistent Infection-style Process Group Membership (free)	The §2 and §3 protocol , including the indirect probe and the refutation mechanism.
Hayashibara et al. — The ϕ Accrual Failure Detector (free)	The §2 detector, from the paper that introduced it.
HashiCorp — Serf and the Lifeguard extensions (free)	SWIM in production , with the modifications needed to make it behave under real-world latency variance.
Chandra & Toueg — Unreliable Failure Detectors for Reliable Distributed Systems (free)	The theory of §1 — what properties a detector must have for consensus to be solvable despite FLP (Module 05).

Module 08 — Distributed Data

Resource	What it covers for this module
Kleppmann — Designing Data-Intensive Applications, chapters 6, 7 and 9	Partitioning, transactions, and 2PC, with the blocking problem of §2 explained properly.
Karger et al. — Consistent Hashing and Random Trees (free)	The §1 algorithm, from the paper that introduced it and founded Akamai.
Shapiro, Preguiça, Baquero & Zawirski — A Comprehensive Study of Convergent and Commutative Replicated Data Types (free)	The CRDT reference. Comprehensive, and the taxonomy in the first sections is the useful part.
Garcia-Molina & Salem — Sagas (free)	The §2 alternative, from 1987 — long before microservices rediscovered it.

Module 09 — Netcode I: The Latency Problem

Resource	What it covers for this module
Glenn Fiedler — Networked Physics and the Gaffer On Games series (free)	The reference for this module and the next two. Start with 'What Every Programmer Needs To Know About Game Networking'.
Valve — Source Multiplayer Networking (free)	A shipping architecture documented by its authors — snapshots, interpolation, prediction, and lag compensation in one coherent description.
Bettner & Terrano — 1500 Archers on a 28.8: Network Programming in Age of Empires (free)	The classic lockstep paper , and still the clearest account of §3's trade-offs and desync hunting.
Ruoyu Sun / GGPO — rollback networking documentation (free)	The rollback architecture, with source.

Module 10 — Netcode II: Prediction and Reconciliation

Resource	What it covers for this module
Glenn Fiedler — Client-Side Prediction and Server Reconciliation (free)	The §1 and §2 technique , with code and with the smoothing detail of the second callout.
Gabriel Gambetta — Fast-Paced Multiplayer (free, interactive)	Interactive demonstrations of prediction, reconciliation, and interpolation , with sliders for latency. The single best way to build intuition for this module.
Valve — Latency Compensating Methods in Client/Server Protocol Design (free)	The original Half-Life paper. Covers §3's interpolation and sets up Module 11.
Infil — The Fighting Game Glossary, and GGPO's documentation (free)	Rollback explained from both the player's and the implementer's side — the §4 human-factors argument is clearest in the player-facing writing.

Module 11 — Trust, Cheating, and Fairness

Resource	What it covers for this module
Valve — Latency Compensating Methods (free)	The §2 rewind technique , documented by the people who popularised it, including the shot-around-a-corner consequence.
Glenn Fiedler — cheat prevention in networked games (free)	The §1 principle, stated as a design rule.
Riot Games — engineering blog on Valorant's netcode and anti-cheat (free)	Server-side visibility (§3) as a deliberate anti-cheat architecture , with the CPU cost discussed openly. The best available account of the information-hiding argument.
Hoglund & McGraw — Exploiting Online Games	The attacker's perspective on §3. Dated in specifics and accurate in structure.

Module 12 — Cloud, Scaling, and Economics

Resource	What it covers for this module
Google — Site Reliability Engineering, chapters on capacity and load (free)	The §3 material , including load shedding and the cascading-failure patterns that scaling can trigger.
Armbrust et al. — A View of Cloud Computing (free)	The §1 economics , argued carefully. Old, and the argument is unchanged.
37signals — the cloud repatriation writeups (free)	The counter-case with real numbers. Read alongside the Berkeley paper rather than instead of it.
Agones, and the major providers' game-server hosting documentation (free)	The §4 workload, handled by tooling built specifically for stateful session placement.

Module 13 — Operating It

Resource	What it covers for this module
Google — Site Reliability Engineering and the SRE Workbook (free)	SLOs, error budgets, and cascading failure — §2 and §3, from the organisation that formalised them. The 'Addressing Cascading Failures' chapter is the one to read first.
Dean & Barroso — The Tail at Scale (free)	The §2 fan-out arithmetic , with hedged and tied requests. Short, and among the most useful papers in systems.
OpenTelemetry and W3C Trace Context (free)	The §1 standard, and the instrumentation libraries for every major language.
FoundationDB — Testing Distributed Systems with Deterministic Simulation (free)	The §3 approach , which is the strongest testing methodology in this subject and is still not widely adopted.
Kyle Kingsbury — Jepsen (free)	Real distributed databases tested against their own claims , with the results. The best available argument for §4's discipline.